



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Business Process Mapping				
Course Code:	BTA1004	Credit Value:	4	Credit Hours:	56
Programs:	BAGC I.T. Business Analysis BTHP I.T. Business Analysis - Hanson				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will model business processes using standard graphical representations, as a flow of activities. Contemporary BPM processes are built to interact as services with other systems, or even to orchestrate or choreograph other systems, including the business processes of other companies. The goal of process modeling is to gain a better understanding of an entire process - not just its components, and ultimately to optimize organizational procedures and their supporting technologies cooperatively. Students will also investigate and use various software products to prepare models that syntactically and logically represent improved business processes.

Date: June 19, 2020

Prepared By: Nathan Abourbih

Approved by: 

BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Document business processes using effective and established research and documentation methods and business improvement procedures. 2. Document business requirements in accordance with standard business practices. 3. Communicate the results of business requirements analysis and documentation to a broad and diverse audience using appropriate communication tools and styles. 4. Apply negotiation and problem-solving skills to build effective stakeholder alignment.
I.T. Business Analysis - Hanson	Ministry Standards 1. Document business processes using effective and established research and documentation methods and business improvement procedures. 2. Document business requirements in accordance with standard business practices. 3. Communicate the results of business requirements analysis and documentation to a broad and diverse audience using appropriate communication tools and styles. 4. Apply negotiation and problem-solving skills to build effective stakeholder alignment.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Business Process Model and Standards
- The Notation in Detail
- Strategic Process Models
- Operational Process Models
- Case Management Model and Notation
- Decision Model and Notation
- Automation

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Trace the history, progression and main contributors to the Business Process Modeling Notation (BPMN) standard.	1.1 Describe the main uses for Business Process Model and Notation (BPMN). 1.2 Trace the origins of BPMN and identify the major contributions of both the Business Process Management Initiative (BPMI) and the Object Management Group (OMG) to the current BPMN 2.0 standard. 1.3 Recognize and describe the four main graphical elements depicted in simple BPMN models: - Flow objects include events, activities, gateways. - Connecting objects include sequence flows, message flows and associations - Swim lanes include pools and lanes - Artifacts include data objects, groups and annotations 1.4 Read, interpret and explain examples of business process diagrams. 1.5 Compare BPMN versions 1.0, 1.1, 1.2, and 2.0 by various attributes such as release date, model types, number of elements, major changes. 1.6 Identify the relationship between BPMN and BPEL (Business Process Execution Language). 1.7 Identify weaknesses of BPMN and what it is not meant to represent. 1.8 List and describe the benefits of adopting the BPMN 2.0 standard in Business Process Management.
2. Relate business process modeling within the context of business process management (BPM).	2.1 Define Business Process Management. 2.2 Assess both automated and non-automated processes from start to finish with the goal of aligning these with business strategy. 2.3 Describe the BPM life cycle: process survey, documentation, analysis, design, implementation and control. 2.4 Explain the role of a process engine and how it contributes to process automation. 2.5 List the customers of a process model and describe their roles played in relation to a project: owner, manager, participant, analyst, engineer. 2.6 Describe a BPMN framework for consulting on projects that support functional (business) and technical (IT) scenarios: - Strategic process models - Operational process models

Learning Outcome	Objectives
3. Apply BPMN (Business Process Mapping Notation) in detail through process analysis.	3.1 Apply the five rules of logical participant(s) in process analysis, by building different models for each participant's perspective. 3.2 Describe by stepping through a simple model the concepts of process instance, token and correlation. 3.3 Describe symbols and their attributes stored in a modeling tool. 3.4 Apply the correct BPMN 2.0 syntax in models as well as practice the BPMN etiquette for representing: 1. Simple tasks and none events 2. Sequence flows 3. Design paths with gateways 3.1 data-based exclusive gateway (XOR), parallel gateway (AND), data-based inclusive gateway (OR), complex gateway 4. Design paths without gateways 5. Lanes 6. Events 6.1 catching, throwing, throwing intermediate, message, timer, error, conditional, signal, terminate, link, compensation, multiple, parallel, escalation, cancel, event-based gateway, event-based parallel gateway 7. Special tasks 7.1 manual, user, service, receive and send, business rule, custom (phone calls, signatures, approvals or rejections, archiving), markers (loops, multiple, compensation), global and call activity 8. Subprocesses 8.1 encapsulate complexity, modularization and reuse, attached events, markers, transactions, event 9. Pools and message flows 9.1 orchestration, rules for application, collaboration, collapse pools, multiple instance pools 10. Data 11. Artifacts 11.1 annotations and groups, custom 3.5 Compare BPMN 2.0 with other notations such as extended event-driven process chain (eEPC), UML (Unified Modeling Language) activity diagram, ibo sequence plan (proprietary to Beratung and Training) for its usability and transferability. 3.6 Interpret choreography and conversation diagrams to quickly recognize critical process points in a collaboration involving a large number of people.
4. Use a case approach to create strategic process models.	4.1 Design strategic (level 1) models that describe a process overview and is intended for fast comprehension by the target audience. 4.2 Review user stories in a case scenario then conduct a thorough process analysis.

Learning Outcome	Objectives
	4.3 Use a restricted symbol set to depict a process model to be communicated at strategic levels. 4.4 Identify the limitations and semantics to convey a process as a conversation at different levels of granularity, as well as conveying a process as a choreography.
5. Define operational process models known as level 2 models.	5.1 Create an extension of level 1 models to show how work is completed by one or more process participants. 5.2 Follow modeling requirements that are syntactically correct and remove semantic irregularities that may be accepted at level 1 models. 5.3 Follow the 6 step procedure for moving level 1 models to level 2 models. 5.4 Describe and present a process model from several participant's point of view. 5.5 Design processes by considering the participants as users of software, and supported by a process engine. 5.6 Identify further requirements for implementing an executable process: functional, non-functional, user interface, data, rules. 5.7 Identify the technical implementation components required beyond the process engine: business logic and rules, screens and screen flows, data transformations 5.8 Apply the FPY (First Pass Yield) approach to the main path of the model to identify the probabilities of correction paths. 5.9 Model explicit errors with event types. 5.10 Identify and list the benefits of subprocesses to determine if generating CMMN is more appropriate to the situation. 5.11 Retrieve business rules from processes to clearly identify business decisions and then use DMN for modeling these.
6. Interpret Case Management Models and Notation (CMMN) as a series of human process flows constrained by rules.	6.1 Identify business situations that use subprocesses (ad hoc, non-interrupting, and event-based) to launch additional tasks that clutter process understanding. 6.2 Contrast CMMN and BPMN standards. Identify three indicators for using CMMN rather than BPMN: 1. No sequence required. 2. Activities dependent on conditions. 3. Dedicated planning phase. 6.3 Interpret a CMMN model consisting of case, tasks, stage, milestones, and sentries symbols. 6.4 Sketch user interface primary elements (task area, context, case status) used for case management scenarios.

Learning Outcome	Objectives
	6.5 Recognize and model CMMN notation elements: Cases and tasks, Sentries, Manual activation rule, Repetition rule, Required rule, Auto complete, Milestones, Stages, Exit criterion, Modeling without do-it class, Events, Case file items, Conditions, Non-blocking tasks, Discretionary tasks. 6.6 Explain how and when BPMN and CMMN models are linked.
7. Design and interpret Decision Model and Notation (DMN) business scenarios.	7.1 Describe the use and application of DMN. 7.2 List typical examples of operational decisions made in businesses that are suitable for automation. 7.3 Describe the DMN specification where decisions depend on decision logic, which is defined by rules, and these can be listed in a decision table. 7.4 Use various DMN elements to create models: simple and complex decision tables using AND / OR conditions, and expressed in Friendly Enough Expression Language (FEEL). 7.5 Describe what the "Hit policy" determines and then apply both the single (unique, first, any, priority), and multiple (collect, rule order, output order) hit policies of the DMN standard. 7.6 Use S-FEEL (Simple FEEL) to express conditions with data types and operators in tables. 7.7 Use the decision requirements graph (DRG) and the decision requirements diagram (DRD) to model decision requirements. 7.8 Create descriptive models that link BPMN and DMN as well as models linking CMMN and DMN. 7.9 Describe the life cycle of decisions.
8. Describe the automation elements involving model execution engines.	8.1 Describe the purpose and benefits of automating business processes and decisions with software. 8.2 Describe how the execution engine functions with respect to the workflow (BPMN), case (CMMN) and decision (DMN) models. 8.3 List several reasons for the adoption of engines in a business. 8.4 Describe the executable process model of a business data model involving service calls and user tasks. 8.5 Describe the challenges when BPM projects adopt classic BPM suites. 8.6 Identify the myth surrounding engine interchangeability. 8.7 Identify business aspects that favor technical process models versus classical software development.

Learning Outcome	Objectives
	8.8 Justify technical challenges of the business environment: 1. Describe benefits of data storage with the process or case and when exceptions must be considered. 2. Handle intermediate events to minimize delays, lost messages, instance terminations. 3. Describe the process testing to ensure automation is supported without compromising the business. 8.9 Describe best practices for introducing a BPM platform.
9. Contrast industry BPMN standards.	9.1 Investigate software tools and products that propose their use of BPMN 2.0, CMMN and DMN. 9.2 Use software to create the three types of business process modeling techniques discussed in the course.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- not applicable

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Presentations
- In-Class Exercises
- Case Studies

Resources Required

Books

Freund, Jakob & Rucker, Bernd, *Real-Life BPMN*, 4th, CreateSpace Independent Publishing
ISBN: 9781086302097

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	50%
Two or more equally weighted projects.	
Applied Activities(G)	30%
Quizzes/Test/Exam	20%

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@CambrianCollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@CambrianCollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://CambrianCollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@CambrianCollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://CambrianCollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Regulations**Lab Attendance**

Lab attendance is mandatory since lab grades include an evaluation of lab reports and lab performance. Lab performance can only be judged in person, requiring that students attend all scheduled labs.